FREELANCE BACKEND ENGINEER

ADRIAN DUMITRU



Java · Spring Boot · Kafka · Kubernetes

Lugano, Switzerland · contract and project work in Switzerland and Northern Italy

AT A GLANCE

Six years building the backbone of enterprise systems

6+

years of backend engineering

Millions

of events handled by the Kafka sidecar I built

100s

of developers building on my microservice templates

3

industries: telecom, energy & utilities, finance

Today I work at Swisscom through adesso, owning a service from solution design to production.

WHAT I OFFER

Three ways I can help your team



Senior backend capacity

I join your team on contract and take ownership of Java/Spring services, from analysis to release.

Day rate, remote or on site



Legacy Java modernization

Step-by-step upgrade of ageing Java systems (Java 8 to 17/21, Spring Boot 3, containers) without stopping the business.

Starts with a fixed-scope assessment



Event-driven integration

Kafka-based integration between systems: topic design, Avro schemas and their evolution, reliable producers and consumers.

Design, build and handover

CASE 1 · ENGINEERING INGEGNERIA INFORMATICA · ENERGY & UTILITIES

A Kafka sidecar at the core of an enterprise event platform

THE CHALLENGE

Thousands of microservices on Kubernetes needed one consistent, reliable way to produce and consume events, without every team re-implementing Kafka.

OUTCOME

Millions of events through one shared component

WHAT I DID

- Designed and built the event sidecar in Java: deployed next to every microservice, it handles all event production and consumption
- Introduced Avro with Schema Registry, managing schema evolution and backward compatibility
- Handled payloads from a few bytes up to about 100 MB
- Re-architected v2 with a colleague: multithreaded pooling and better memory management for lower latency under load

STACK

Java · Kafka · Avro · Schema Registry · Kubernetes

CASE 2 · ENGINEERING INGEGNERIA INFORMATICA

Microservice templates adopted by hundreds of developers



THE CHALLENGE

Each new microservice had to be wired to the same databases, secret stores and platform services before any business logic could be written.

OUTCOME

Teams start from business logic, not plumbing

WHAT I DID

- Built reusable Spring Boot templates, pre-wired to the platform's databases and services
- Credential resolution from AWS and Azure secret stores; connectors for PostgreSQL, HBase, DynamoDB and InfluxDB
- Extended the approach to .NET templates and an Envoy WASM filter written in Rust
- Supported other teams deploying their services on Kubernetes

STACK

Spring Boot · Quarkus · .NET · Rust · AWS · Azure · Kubernetes

CASE 3 · SWISSCOM (VIA ADESSO) · TELECOM

Automatic network discovery for a national telecom operator



THE CHALLENGE

Network devices had to be found, inventoried and set up for metrics collection. Doing that by hand does not scale.

OUTCOME

Owned end to end, from design to production

WHAT I DID

- Designed and developed a Spring Boot service that discovers devices via SNMP (v2c and v3) and inventories them
- Automatic generation of Telegraf configuration files for metrics collection
- Concurrent SNMP polling with SNMP4J, with careful handling of SNMPv3 security (USM, engine ID)
- Wrote the solution designs and carry out the production deployments myself

STACK

Java · Spring Boot · SNMP4J · Telegraf · Kafka · Kubernetes

CASE 4 · HORSA · CONTRACT ENGAGEMENT, 2023

Recovering an undocumented legacy system and migrating its database



THE CHALLENGE

A production system with no version control, no documentation and nobody left who knew it, plus a database that had to move to another vendor.

OUTCOME

A black box made understandable and migrated

WHAT I DID

- Reconstructed the source code by decompiling the production JAR
- Reverse-engineered the data model and the business rules
- Delivered a full cross-vendor database migration between MySQL and Oracle
- Debugged and fixed production issues in Spring and Spring Boot services

STACK

Java · Spring · MySQL · Oracle · Docker

CASE 5 · OPEX · CO-FOUNDER (OWN VENTURE)

A fintech back end built for compliance from day one

THE CHALLENGE

A financial platform for freelancers connects to their bank accounts, so it has to meet PSD2 and GDPR requirements.

OUTCOME

Security and compliance designed in, not bolted on

WHAT I DID

- Designed the back-end architecture with Spring Boot, Keycloak and PostgreSQL
- Open banking integration through Salt Edge (PSD2)
- Strong customer authentication with TOTP; AES-256-GCM encryption of sensitive data
- GDPR consent tracking and audit logging; invoicing, cash-flow and quarterly VAT forecasting

STACK

Java · Spring Boot · Keycloak · PostgreSQL · PSD2 · GDPR

SKILLS IN DEPTH

What I have actually done with each technology

Area	What I have done	Where
Java & Spring Boot	Designed and built production services, both new and on legacy Java 8 code	Swisscom, Engineering, contracts
Kafka & event-driven	Built a platform-wide sidecar; Avro, Schema Registry, schema evolution	Engineering, Swisscom
Kubernetes & cloud	Deploying and running services on Kubernetes; AWS Lambda, IAM, Secrets Manager; Azure	Engineering, Prometeia, Swisscom
Legacy & reverse engineering	Decompiling, production debugging, MySQL ↔ Oracle migration	Horsa, 2023
Security & compliance	OAuth2 PKCE end to end, Keycloak, TOTP authentication, encryption, GDPR	Prometeia, Opex
Networking	SNMP v2c/v3 discovery, device inventory, Telegraf metrics	Swisscom
Delivery	Solution design, production releases, support to other teams	Swisscom, Engineering

HOW I WORK

From the first analysis to production



01

Understand

I read the code and talk to the people who run it before proposing anything.

02

Design

A written solution design, agreed with you, with risks and alternatives.

03

Build

Small, reviewable increments, tested and documented.

04

Ship & hand over

Production deployment, then documentation and knowledge transfer to your team.

I am used to working embedded in client teams, with their processes and tools.

WORKING TOGETHER

Practical details



Where

Remote across Switzerland and Italy · on site in Ticino · hybrid in Zurich or Milan



Languages

Italian and Romanian (native) · English (B1/B2)



Formats

Day-rate contract · fixed-scope assessment for legacy Java systems



Availability

Start date to be agreed

Let's talk

Happy to have a 20-minute call to understand what you need.



dumitru.adrian.dev@gmail.com

+41 79 478 68 22

linkedin.com/in/adrian-dumitru-dev

github.com/Adryd99

Lugano, Ticino